

PELLO, Finland

WMO: 028440

Lat: 66.773N

Lon: 23.963E

Elev: 282

StdP: 14.55

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.4	-18.3	-28.5	1.1	-23.3	-23.3	1.5	-18.3	18.4	24.6	16.3	24.5	2.3	320	0.668

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.3	77.9	62.2	74.3	60.3	70.8	59.0	65.0	73.6	62.8	70.6	60.8	67.6	7.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.9	83.6	68.0	59.8	77.6	65.3	57.9	72.3	63.6	30.2	73.8	28.5	70.3	27.1	67.6	72.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.5	14.3	12.5	DB	-28.9	82.0	5.1	4.0	-32.6	84.8	-35.6	87.2	-38.5	89.4	-42.2	92.3	(4)
(5)			WB	-28.4	67.5	5.2	2.5	-32.1	69.3	-35.1	70.8	-38.0	72.2	-41.8	74.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	34.0	10.0	11.0	20.8	32.6	44.2	54.6	60.5	56.1	46.2	33.7	22.0	15.2
	DBStd	20.02	14.02	14.33	10.37	7.30	7.67	6.52	5.64	6.03	6.47	8.66	11.40	13.99
	HDD50	6593	1240	1093	906	523	214	28	1	15	149	506	839	1079
	HDD65	11336	1705	1513	1371	973	645	318	162	282	563	971	1289	1544
	CDD50	765	0	0	0	0	35	164	326	204	36	0	0	0
	CDD65	33	0	0	0	0	0	5	22	6	0	0	0	0
	CDH74	346	0	0	0	0	15	64	214	53	0	0	0	0
	CDH80	43	0	0	0	0	1	8	31	3	0	0	0	0
Wind	WSAvg	5.7	5.1	5.4	6.1	6.1	6.5	6.2	5.8	5.4	5.5	5.5	5.1	5.3
Precipitation	PrecAvg	21.0	1.5	1.2	1.1	1.1	1.4	2.4	3.0	2.5	1.9	1.7	1.9	1.6
	PrecMax	26.8	2.6	2.6	2.3	2.4	3.9	4.4	5.0	5.0	3.6	3.7	3.3	3.1
	PrecMin	15.7	0.3	0.2	0.2	0.3	0.2	0.3	0.8	0.6	0.5	0.2	0.5	0.3
	PrecStd	3.6	0.5	0.6	0.6	0.6	0.7	1.0	1.2	1.1	0.9	0.8	0.8	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.0	41.1	45.9	58.5	75.8	80.3	83.7	79.5	65.7	53.2	44.1	40.1
		MCWB	36.0	36.7	38.6	46.0	56.5	62.4	65.1	65.9	56.8	49.8	42.3	36.8
	2%	DB	34.0	35.1	41.7	51.8	69.4	74.8	79.4	74.6	62.0	50.5	40.5	36.3
		MCWB	32.5	33.1	35.7	42.1	54.6	58.8	63.2	61.4	54.4	48.1	38.9	34.9
	5%	DB	32.0	32.8	38.0	48.0	62.6	71.0	75.8	70.5	59.2	48.1	37.4	34.1
		MCWB	31.5	31.8	33.2	40.0	50.3	56.9	62.0	60.3	54.0	46.0	36.6	33.2
	10%	DB	29.3	30.6	34.5	44.1	58.0	67.0	71.9	66.8	56.4	45.2	35.2	32.7
		MCWB	28.7	29.7	31.6	37.2	48.0	55.1	60.2	58.7	52.5	43.3	34.5	32.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.3	37.5	39.2	47.2	58.9	64.8	68.1	67.4	59.1	51.2	42.9	37.4
		MCDB	38.5	40.8	45.7	57.8	71.7	77.1	76.7	75.8	62.5	52.0	43.7	38.9
	2%	WB	33.0	33.5	36.1	43.0	55.4	61.3	65.7	64.3	56.7	48.7	39.0	35.4
		MCDB	33.8	34.6	40.8	50.1	65.9	70.3	74.3	70.7	59.6	50.0	40.0	36.4
	5%	WB	31.4	31.9	33.9	40.5	52.0	58.7	63.8	61.7	55.0	46.1	36.7	33.3
		MCDB	32.1	32.7	36.5	47.4	61.3	67.5	72.5	67.6	58.1	47.9	37.4	33.9
	10%	WB	28.6	29.6	32.2	38.0	48.8	56.7	61.9	59.7	52.9	43.5	34.6	32.2
		MCDB	29.3	30.6	34.4	43.2	56.5	64.8	69.6	65.1	56.0	45.1	35.1	32.8
Mean Daily Temperature Range	5% DB	MDBR	15.2	15.1	18.6	16.8	16.6	16.0	16.3	16.2	14.2	10.1	10.4	13.2
		MCDBR	12.1	11.5	18.7	22.0	25.4	22.8	22.6	21.7	17.5	10.9	8.4	10.2
		MCWBR	11.5	10.2	13.8	13.4	13.0	9.7	9.1	10.5	10.8	8.8	7.9	9.5
	5% WB	MCDBR	12.2	11.4	15.5	20.5	22.5	18.5	18.4	17.9	13.9	9.4	7.9	10.1
		MCWBR	11.7	10.3	12.0	12.8	12.2	9.3	8.9	9.5	9.2	8.2	7.6	9.7
Clear-Sky Solar Irradiance	taub	0.197	0.263	0.296	0.329	0.342	0.346	0.357	0.341	0.314	0.286	0.344	N/A	
	taud	1.936	2.107	2.181	2.247	2.447	2.468	2.453	2.483	2.540	2.357	2.668	N/A	
	Ebn at Noon	59	180	238	262	272	274	266	259	240	178	66	0	
	Edh at Noon	4	16	26	32	30	30	30	26	19	13	5	0	
All-Sky Solar Radiation	RadAvg	21	178	643	1199	1515	1639	1541	1125	602	212	40	0	
	RadStd	1	29	60	79	150	126	147	93	75	39	5	0	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.26	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (45 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page